DHVANI Shield v1 – Complete KiCad Guide

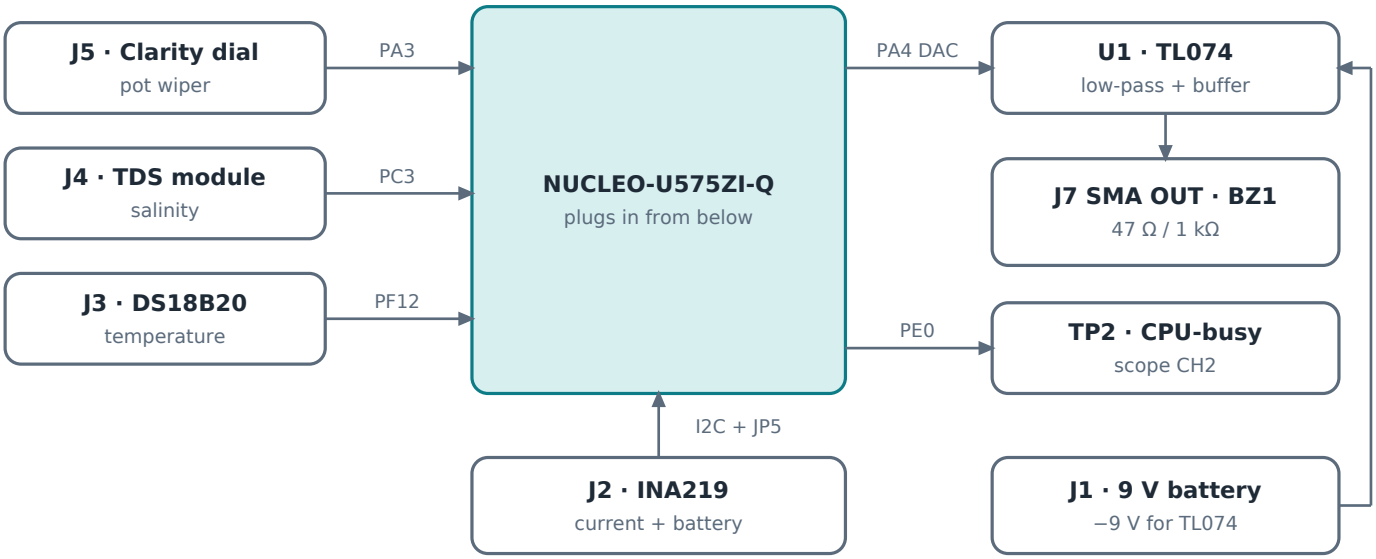
PCB that plugs onto the NUCLEO-U575ZI-Q and replaces the breadboard · SIH 2026 · PS 26058 · for video and roadmap only (not fabricated)

1. What you are making

A small board (a “shield”) that sits on top of the Nucleo. It holds everything that is on the breadboard today: the TL074 filter, the connectors for all sensors, the buzzer and an SMA connector for the oscilloscope.

- **Same circuit as the breadboard.** No new parts, no pin changes.
- **Tool:** KiCad (any recent version).
- **Board:** 2 layers, 1.6 mm thick.
- **Honesty:** label it “Phase 2: designed, not yet fabricated”. Do not say it fits other Nucleo boards.

2. Block diagram



5 V and 3.3 V come from the Nucleo headers (CN8). All grounds are joined.

3. What the SMA output should give

The SMA connector replaces the old scope test point. Signal path: **TL074 pin 8 → R3 (47 Ω) → SMA centre pin**. The SMA outer shell goes to **GND**. All values below were measured on the real board.

Item	Value
Waveform	Short bursts (pings) with silence between them. Inside each ping: a chirp (LFM or geometric sweep) or a Barker-13 coded pulse, with a smooth swell-and-fade shape.
Frequency – audible mode	2–17 kHz in clear water, 2–5.8 kHz in muddy water. Coded (BPSK) pulse peaks near 8.7 kHz.
Frequency – ultrasonic mode	About 100–250 kHz (measured peaks: 110 kHz muddy, 179 kHz clear).
Pulse length	About 4 ms (clear) to 15–16 ms (muddy).
Amplitude	1.90 Vpp (fresh water) to 2.92 Vpp (salty water).

Item	Value
DC offset	About 1.65 V. The signal swings around 1.65 V (roughly 0.2 V to 3.1 V at most).
Filter	Low-pass, cut-off about 340 kHz. Removes the DAC “staircase” and its 890 kHz image.
Output resistance	47 Ω (R3).

Oscilloscope setting

- Use **1 MΩ (high impedance)** input. All numbers above were measured this way.
- **Do not use 50 Ω.** With 50 Ω the 47 Ω resistor and the 50 Ω input split the signal, so you see about half (≈ 1.5 Vpp), and the TL074 cannot push that much current cleanly, so the wave distorts.
- To see the shape: **DC coupling**. For the FFT: **AC coupling** (removes the 1.65 V offset).

4. Parts list with KiCad symbols and footprints

Ref	Part / value	KiCad symbol	Footprint (suggestion)
U1	TL074 quad op-amp	`Amplifier_Operational:TL074`	`Package_DIP:DIP-14_W7.62mm`
R1, R2	3.3 kΩ (filter)	`Device:R`	`Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal`
R3	47 Ω (SMA output)	`Device:R`	same as above
R4	1 kΩ (buzzer)	`Device:R`	same as above
R5	4.7 kΩ (DS18B20 pull-up)	`Device:R`	same as above
C1	200 pF (or 2 × 100 pF)	`Device:C`	`Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm`
C2	100 pF	`Device:C`	same as above
C3, C4	100 nF (supply decoupling)	`Device:C`	same as above
J7	SMA jack (scope output)	`Connector:Conn_Coaxial`	search “SMA” in Connector_Coaxial (edge or vertical)
BZ1	Buzzer	`Device:Buzzer`	search “Buzzer” in Buzzer_Beeper (12 mm)
J1	9 V battery, 2-pin	`Connector_Generic:Conn_01x02`	`Connector_JST:JST_PH_B2B-PH-K_1x02_P2.00mm_Vertical`
J2	INA219, 6-pin	`Connector_Generic:Conn_01x06`	`Connector_PinHeader_2.54mm:PinHeader_1x06_P2.54mm_Vertical`
J3	DS18B20, 3-pin screw	`Connector_Generic:Conn_01x03`	search “TerminalBlock” (3-pin, 5.08 mm)
J4, J5	TDS / dial, 3-pin	`Connector_Generic:Conn_01x03`	`Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical`
J6	JP5 link, 2-pin	`Connector_Generic:Conn_01x02`	`Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical`
TP2, TP3	CPU-busy, GND	`Connector:TestPoint`	`TestPoint:TestPoint_Pad_D1.5mm`
CN7	Nucleo header 2×10	`Connector_Generic:Conn_02x10_Odd_Even`	2.54 mm 2×10 pin header
CN8	Nucleo header 2×8	`Connector_Generic:Conn_02x08_Odd_Even`	2.54 mm 2×8 pin header

Ref	Part / value	KiCad symbol	Footprint (suggestion)
CN9	Nucleo header 2×15	`Connector_Generic:Conn_02x15_Odd_Even`	`2.54 mm 2×15 pin header`
CN10	Nucleo header 2×17	`Connector_Generic:Conn_02x17_Odd_Even`	`2.54 mm 2×17 pin header`

Footprint names can differ a little between KiCad versions. If one is not found, type the keyword (SMA, TerminalBlock, Buzzer) in the footprint search.

5. Connections (draw the schematic from this)

Net name	Nucleo pin	Goes to on the shield
DAC_OUT	CN7-9 (PA4)	R1 → node X → R2 → U1 pin 5
Filter caps	—	X → C1 → U1 pin 7; U1 pin 5 → C2 → GND
Filter feedback	—	U1 pin 6 ↔ U1 pin 7
Buffer	—	U1 pin 7 → U1 pin 10; U1 pin 9 ↔ U1 pin 8
OUT_SMA	—	U1 pin 8 → R3 (47 Ω) → J7 centre; J7 shell → GND
OUT_BUZZ	—	U1 pin 8 → R4 (1 kΩ) → BZ1 → GND
+5V	CN8-9	U1 pin 4; C3 from pin 4 to GND
−9V	J1 (battery −)	U1 pin 11; C4 from pin 11 to GND; battery + → GND
Unused amps A, D	—	pin 3 → GND, pin 2 ↔ pin 1; pin 12 → GND, pin 13 ↔ pin 14
+3V3	CN8-7	J2 VCC, J3 VDD, J4 +, J5 top pin, R5
GND	CN8-11, CN7-8	all grounds, TP3, J7 shell, ground zone
TEMP_DATA	CN7-20 (PF12)	J3 DATA; R5 from DATA to +3V3
TDS	CN9-5 (PC3)	J4 signal (A)
CLARITY	CN9-1 (PA3)	J5 middle pin (wiper)
I2C_SDA	CN7-4 (PB9)	J2 SDA
I2C_SCL	CN7-2 (PB8)	J2 SCL
I_SENSE	JP5 pins 1 and 2	J6 → J2 VIN+ / VIN−
CPU_BUSY	CN10-33 (PE0)	TP2

> **Nucleo header positions:** do not guess them. Take the CN7–CN10 positions from ST's official NUCLEO-144 board drawing (user manual UM2179), or measure the real board with a caliper. Check that pin 1 of each header is on the correct side – one flipped header breaks every connection.

6. Step by step in KiCad

A. Schematic

1. Create a new project: **DHVANI_Shield_v1**.
2. Open the Schematic Editor. Add every part from section 4 (shortcut **A**).
3. Wire them using section 5. Use **net labels** (shortcut **L**) like DAC_OUT, I2C_SDA – cleaner than long wires.
4. Add power symbols +5V, +3V3, −9V and GND.

5. Add a **PWR_FLAG** on +5V, +3V3, −9V and GND (they come in through connectors, so KiCad needs this).
6. Run **Inspect → Electrical Rules Checker**. Fix until there are 0 errors.
7. Assign footprints: **Tools → Assign Footprints** (use section 4).

B. Board setup (File → Board Setup)

Setting	Value
Copper layers	2 (F.Cu = signals, B.Cu = ground)
Board thickness	1.6 mm
Signal track width	0.25 mm
Power track width (+5V, +3V3, −9V)	0.5 mm (make a “Power” net class)
Clearance	0.2 mm
Via	0.6 mm pad, 0.3 mm drill

C. PCB layout

1. **Tools → Update PCB from Schematic** (F8).
2. Draw the board outline on **Edge.Cuts**, sized to the Nucleo-144 board.
3. Place CN7–CN10 at the exact Nucleo positions (see the note above), on the **bottom side** so the shield plugs down onto the Nucleo.
4. Place the **filter group** (U1, R1, R2, C1, C2) close together, near the PA4 pin.
5. Place **C3 and C4 right next to U1 pins 4 and 11**.
6. Put **J7 (SMA) on the board edge** with **R3 right beside it**; keep the trace from U1 pin 8 short.
7. Put all connectors (J1–J6) on the edges; TP2/TP3 at the front edge.
8. Route signals first, then power (0.5 mm).
9. Add a **GND zone on B.Cu** (shortcut Ctrl+Shift+Z), add ground vias around the SMA and around the filter.
10. Keep the I2C and temperature wires away from the filter area.
11. Run **Inspect → Design Rules Checker** until there are 0 errors.

D. Silkscreen (F.Silkscreen layer)

- Title: **DHVANI Shield v1 · SIH 2026 · PS 26058**
- Labels next to connectors: **SMA OUT, DIAL, TDS, TEMP, INA219, 9V, CPU-BUSY, GND**

E. 3D view

- **View → 3D Viewer** (Alt+3). Check every part has a 3D model and nothing overlaps.

7. What to send back (for the video and PPT)

Deliverable	How to make it	Used for
Schematic image	File → Plot (or screenshot), 1920 × 1080 PNG	Video 2 s + PPT
PCB top view image	PCB Editor screenshot, 1920 × 1080 PNG	Video 2 s + PPT
3D render video	Screen-record the 3D Viewer rotating, 8–10 s	Video 2 s
KiCad project folder	Zip the whole folder	Backup / judges

> **Caption for video and slide:** “Phase 2: integrated DHVANI shield, replaces the breadboard (designed, not yet fabricated).”

8. Final checklist

- ERC: 0 errors. DRC: 0 errors.
- Every net in section 5 is connected.
- Nucleo header positions taken from the ST drawing, pin 1 correct.
- SMA at the edge, R3 next to it, ground vias around it.
- C3/C4 touching U1 power pins.
- Silkscreen title and connector labels added.
- 3D view looks clean; images and video exported.